Chapter 2: Starting at the Beginning: the Natural Numbers

Analysis I, by Terrence Tao

A: The Peano Axioms

I include here the five Peano Axioms for easy reference. Please note that $n{+}{+}$ is the successor function and may also be denoted as $S(n)$. In general, I use $n{+}{+}$ if it was mentioned in the definition or for a problem. Then, I use $S(n)$ for when I am showing the proof since the grouping of terms are more clear.

Axiom 1. *0 is a natural number*

Axiom 2. *If $n \in \mathbb{N}$, then $n{+}{+} \in \mathbb{N}$.*

Axiom 3. *For all $n \in \mathbb{N}$, $n{+}{+} \neq 0$.*

Axiom 4. *(Injectivity of the Successor Function) Let $n, m \in \mathbb{N}$. If $n{+}{+} = m{+}{+}$, then $n = m$.*

Axiom 5. *(Principle of Mathematical Induction). Let $P(n)$ be any property pertaining to a natural number n. Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true $P(n{+}{+})$ is also true. Then, $P(n)$ is true for every natural number n.*

B: Addition

Definition 1 - Addition of Natural Numbers

Let $m, n \in \mathbb{N}$. To add zero to m, we define $0 + m := m$. Now, suppose inductively that we have defined how to add n to m, so we know what $n + m$ means. Then, we can add $n{+}{+}$ to m by defining $(n{+}{+}) + m = (n + m){+}{+}$.

Problem 1 - Associativity of Addition

For any natural numbers a, b, c, we have $(a + b) + c = a + (b + c)$.

Proof. Keep b and c fixed and induct on a. We begin with the base case of $a = 0$, so $(0 + b) + c = b + c$. Notice also that $0 + (b + c) = b + c$. By the transitivity of equality, $(0 + b) + c = b + c = 0 + (b + c)$. Hence, $(0 + b) + c = 0 + (b + c)$.

Now, suppose inductively that $(a + b) + c = a + (b + c)$. We'll begin from the right hand to show that that addition is still associative for $S(a)$, so notice:

$$\begin{aligned}
S(a) + (b + c) &= S\big(a + (b + c)\big) && \text{by Definition 1} \\
&= S\big((a + b) + c\big) && \text{by the Inductive Hypothesis} \\
&= S(a + b) + c && \text{by Definition 1} \\
&= \big(S(a) + b\big) + c && \text{by Definition 1.}
\end{aligned}$$

$\square$

Problem 2 - Unique Predecessor of Positive Natural Numbers

For any positive natural number a, there exists a unique natural number b such that $b{+}{+} = a$.

Proof. We prove this by induction on a. For the base case of $a = 0$, this is vacuously true since a is not a positive natural number. Now, suppose inductively that there exists a unique predecessor b such that $b{+}{+} = a$. We don't actually need the inductive hypothesis. I am just stating it for formality. However, notice that there exists a predecssor for $a{+}{+}$. It is just a. This is also unique by the injectivity of the successor function. $\square$

Lemma 1. *Let $a, b, k \in \mathbb{N}$. If $a = b + k$, THEN $a = b$ if and only if $k = 0$.*

Proof. Suppose $a = b+k$. Assume $a = b$. Substituting a for b gives us $a = b+k = a+k$. By the cancellation law, we have $a = a + k$ implying $k = 0$. Conversely, suppose $k = 0$. Trivially, $a = b + k = b + 0 = b$. $\square$

Definition 2 - - Ordering of the Natural Numbers

Let $n, m \in \mathbb{N}$. We say $n \geq m$ or $m \leq n$ if and only if $n = m + a$ for some $a \in \mathbb{N}$. We say $n > m$ or $m < n$ if and only if $n \geq m$ and $n \neq m$.

Problem 3 - Properties of Order for Natural Numbers

Show that the properties of order for the natural numbers are true. Let $a, b, c \in \mathbb{N}$. Then, the following properties hold:

Proposition 3.1. *(Order is reflexive) $a \geq a$*

Proof. Notice $a = a + 0$. Since $0 \in \mathbb{N}$, $a \geq a$. □

Proposition 3.2. *(Order is transitive) If $a \geq b$ and $b \geq c$, then $a \geq c$.*

Proof. Suppose $a \geq b$ and $b \geq c$, so $a = b + k$ and $b = c + l$ for some $k, l \in \mathbb{N}$. Then, we can do substitution to get $a = (c + l) + k = c + (l + k)$. Since $(l + k) \in \mathbb{N}$, $a \geq c$. □

Proposition 3.3. *(Order is anti-symmetric.) If $a \geq b$ and $b \geq a$, then $a = b$.*

Proof. Suppose $a \geq b$ and $b \geq a$, so $a = b + k$ and $b = a + l$ for some $k, l \in \mathbb{N}$. Substiting b for $a + l$ gives us $a = (a + l) + k$. By the cancellation law, we get $0 = l + k$. Hence, $l = k = 0$. Because □

Proposition 3.4. *(Addition preserves order.) We have that $a \geq b$ if and only if $a + c \geq b + c$.*

Proof. For the forward direction, suppose $a \geq b$, so $a = b+k$ for some $k \in \mathbb{N}$. Then, $a+c = (b+k)+c = (b+c)+k$. Hence, $a+c \geq b+c$. Conversely, suppose $a + c \geq b + c$. Then, $(a + c) = (b + c) + k$ for some $k \in \mathbb{N}$. By the cancellation law, we have $a = b + k$, which implies $a \geq b$. □

Proposition 3.5. *(Relationship between $\geq$ and $>$.) We have $a < b$ if and only if $a{+}{+} \leq b$*

Proof. For the forward direction, suppose $a < b$ (i.e., $b > a$), so $b = a + k$ for some $k \in \mathbb{N}$ and $b \neq a$. By Lemma 1, we have that since $b \neq a$, $k \neq 0$. Since k is a postive number, there exists a unique natural number l such that $S(l) = k$. Then, $a + k = a + S(l) = S(a + l) = S(a) + l = b$. Hence, $S(a) \leq b$.

Conversely, suppose $S(a) \leq b$ (i.e., $b \geq S(a)$), so there exists a $j \in \mathbb{N}$ such that $b = S(a) + j$. Notice $b = S(a) + j = a + S(j)$. Since $S(j) \neq 0$, we have that $b \neq S(a)$ by Lemma 1. Therefore, $a < b$. □

Proposition 3.6. *(Another Relationship between $\geq$ and $>$.) We have $a < b$ if and only if $b = a + d$ for some positive natural number d.*

Proof. For the forward direction, suppose $a < b$ (i.e., $b > a$), so $b = a + d$ for some $d \in \mathbb{N}$. and $b \neq a$. By Lemma 1, we have that since $b \neq a$, $d \neq 0$. Therefore, d is a positive natural number.

Conversely, suppose $b = a + d$ for some positive natural number d. By Lemma 1, we have that since $d \neq 0$, $b \neq a$. Hence, $a < b$.

□

Problem 4 - Propositions to Prove the Trichotomy of Order

Justify the three statements marked 'why?'. For the three statements, let $a, b \in \mathbb{N}$.

Proposition 4.1. *For any b, $0 \leq b$.*

Proof. We induct on b. For the base case of $b = 0$, $0 \leq 0$ since $0 = 0 + 0$. Now, suppose inductively that $0 \leq b$, so $b = 0 + b$. Notice that $S(b) = S(0+b) = 0+S(b)$. Therefore, $0 \leq S(b)$. □

Proposition 4.2. *If $a > b$, then $a{+}{+} > b$.*

Proof. Suppose $a > b$, so $a = b+k$ for some positive natural number k. Then, $S(a) = S(b+k) = b+S(k)$. Since $S(k) \neq 0$ by Axiom 3, $S(k)$ is a positive natural number. Hence, $S(a) > b$. □

Proposition 4.3. *If $a = b$, then $a{+}{+} > b$.*

Proof. Suppose $a = b$. Then, $b = 0+b$, so $a = 0+b$. Notice $S(a) = S(0+b) = S(0)+b = b+S(0)$. Since $S(0) \neq 0$, $S(a) > b$. □

Problem 5 - Strong Principle of Induction

Let $m_0 \in \mathbb{N}$, and let $P(m)$ be a property pertaining to an arbitrary $m \in \mathbb{N}$. Suppose that for all $m \geq m_0$, we have the following implication:

$$\forall m' \in [m_0, m),\ P(m') \text{ is true} \Longrightarrow P(m) \text{ is true.}$$

Then, we can conclude that $P(m)$ is true for all natural numbers $m \geq m_0$.

Proof. To make what we are proving more clear, we construct a new property $Q(n)$ which has that:

$$Q(n) : P(m) \text{ is true for all } m \in [m_0, n)$$

For the base case, $Q(0)$ is true because there does not exist any $m' \in [m_0, 0)$. Hence, $Q(0)$ is vacuously true. We now consider two cases: $n \leq m_0$ and $n > m_0$.

1. Say $n < m_0$. Then, the set $[m_0, n)$ is empty, so $Q(n)$ is vacuously true. Notice also that $n{+\!+} \leq m_0$, which means that $[m_0, n{+\!+})$ is still empty. Hence, $Q(n{+\!+})$ is also vacuously true.

2. Say instead that $n \geq m_0$. This is the more interesting case, and so we will have to actually start inducting. Suppose that $Q(n)$ is true. Hence, for all $m' \in [m_0, n),\ P(m')$ is true. By the implication presented in the problem, $P(n)$ must also be true. Because $P(m_0)$ up to $P(n-1)$ is true by the inductive hypothesis and then $P(n)$ is true as proven, we have shown that for all $m' \in [m_0, n{+\!+}),\ P(m')$ is true. This means that $Q(n{+\!+})$ is also true.

□

Problem 6 - Principle of Backwards Induction

Let n be a natural number, and let $P(m)$ be a property pertaining to the natural numbers such that $P(m{+\!+})$ being true implies $P(m)$ being true. Suppose that $P(n)$ is also true. Prove that $P(m)$ is true for all natural numbers $m \leq n$. This is known as the principle of backwards induction.

Proof. This is the informal version that Gemini said was mid. Let $m \in \mathbb{N}$. Suppose that $m \leq n$, so $n = m + a$ for some $a \in \mathbb{N}$. Then, to prove that $P(m)$ is true, use the implication of $P(m{+\!+}) \implies P(m)$ presented in the problem a times. Start with that $P(n)$ is true. If $n = 0$, then there is no number less than it do backwards induction. Otherwise, there exists a unique natural number l such that $l{+\!+} = n$. Hence, $P(n) = P(l{+\!+}) \implies P(l)$. This can be done a times to eventually reach that $P(m)$ is true. □

Proof. This is the more formal version. Let us define a new property $Q(a)$ such that

$$Q(a) : \text{If } n = m + a \text{ for some } m \in \mathbb{N}, \text{ then } P(m) \text{ is true.}$$

For the base case of $a = 0$, this would mean that $n = m + 0 = m$. Since we assumed that $P(n)$ is true, $P(m)$ must also be true. Now, assume inductively that $Q(a)$ is true. Now, we will show that $Q(a{+\!+})$ is true. Suppose $n = m + S(a)$, so $n = S(m) + a$ with $S(m) \in \mathbb{N}$. Since $Q(a)$ is true, $P(m{+\!+})$ is true. With the core impplication of the problem that $P(m{+\!+}) \implies P(m)$, we have that $P(m)$ must also be true. Hence, $Q(a{+\!+})$ is true. □

C: Multiplication

Definition 3 - Multiplication of Natural Numbers

Let $m, n \in \mathbb{N}$. To multiply zero to m, we define $0 \times m := 0$. Now, suppose inductively that we have defined how to multiply n to m, so we know what $n \times m$ means. Then, we can multiply $n{+}{+}$ to m by defining $(n{+}{+}) \times m := (n \times m) + m$.

Problem 1 - Commutativity of Multiplication

Let $n, m \in \mathbb{N}$. Then, $n \times m = m \times n$.

For the proof, we need to do two simpler propositions, which will be helpful for proving Problem 1.

Proposition 1.1. *We have* $n \times 0 = 0$.

Proof. We induct on n. Suppose the base case of $n = 0$. By the definition of multiplication, $0 \times 0 = 0$. Now, suppose inductively that $n \times 0 = 0$. Notice $S(n) \times 0 = (n \times 0) + 0 = 0 + 0 = 0$. □

Proposition 1.2. *We have* $m \times S(n) = (m \times n) + m$.

Proof. We induct on m while keeping n fixed. Suppose the base case of $m = 0$. Then, the LHS evaluates to 0 because $0 \times S(n) = 0$ by definition of multiplication. The RHS evaluates also to 0 because $(0 \times n) + 0 = 0 + 0 = 0$. Hence, the LHS equals to the RHS.

Now, suppose inductively that $m \times S(n) = (m \times n) + m$. Notice:

$$\begin{aligned}
S(m) \times S(n) &= m \times S(n) + S(n) && \text{by Definition 3} \\
&= (m \times n) + m + S(n) && \text{by the inductive hypothesis} \\
&= (m \times n) + S(m) + n && \text{by manipulating the successor function} \\
&= (m \times n) + n + S(m) && \text{by commutativity of addition} \\
&= S(m) \times n + S(m) && \text{by Definition 3}
\end{aligned}$$

□

We now begin the actual proof for Problem 1.

Proof. We induct on n while keeping m fixed. For the base case $n = 0$, the LHS equals to 0 because $0 \times m = 0$ by the definition of multiplication. The RHS equals to 0 because $m \times 0 = 0$ by the first proposition.

Now, suppose inductively that $n \times m = m \times n$. Notice:

$$\begin{aligned}
S(n) \times m &= (n \times m) + m && \text{by Definition 3} \\
&= (m \times n) + m && \text{by the inductive hypothesis} \\
&= m \times S(n) && \text{by second proposition}
\end{aligned}$$

□

Problem 2 - Positive numbers having no zero divisors

Let $n, m \in \mathbb{N}$. Then, $n \times m = 0$ if and only if $n = 0$ or $m = 0$.

Proof. To prove the forward direction, we induct on n while keeping m fixed. To make the induction we clear, we formulate the property $P(n)$, which is just the contrapositive of the forward direction, as:

$$P(n, m) : \text{If } n \neq 0 \text{ and } m \neq 0, \text{ then } n \times m \neq 0.$$

For the base case, $P(0, m)$ is vacuously true. Now, suppose $P(n, m)$ is inductively true. We will show that $P(n{+}{+}, m)$ is also true. Notice that $S(n) \times m = (n \times m) + m$ Since $n \times m \neq 0$ (and as such is a positive number), $S(n) \times m \neq 0$ (and is thus also a positive number.)

Conversely, suppose $n = 0$ or $m = 0$. We consider two cases. Say $n = 0$. Then, $n \times m = 0 \times m = 0$ by Definition 3. Say instead that $m = 0$. Then, $0 \times n = n \times 0 = 0$ by the commutativity of multiplication and Definition 3. In either case, $n \times m = 0$. □

Problem 3 - Associativity of Multiplication

For any natural numbers a, b, c, we have $a \times (b \times c) = (a \times b) \times c$.

Proof. We induct on a while keeping b and c fixed. For the base case $a = 0$, we have for the LHS that $0 \times (b \times c) = 0$. For the RHS, we have $(0 \times b) \times c = 0 \times c = 0$. The LHS equals the RHS.

Now, suppose inductively that $a \times (b \times c) = (a \times b) \times c$. Notice:

$$\begin{aligned} S(a) \times (b \times c) &= \big(a \times (b \times c)\big) + (b \times c) \\ &= \big((a \times b) \times c\big) + (b \times c) \\ &= \big((a \times b) + b\big) \times c \\ &= \big(S(a) \times b\big) \times c \end{aligned}$$

□

Problem 4 - Identity

Prove the identity $(a + b)^2 = a^2 + 2ab + b^2$ for all natural numbers a, b.

Proof. We induct on a while keeping b fixed. For the base case $a = 0$, observe that for the LHS, $(0 + b)^2 = b^2$. While for the RHS, $0^2 + 2(0)b + b^2 = b^2$. Hence, the LHS equals the RHS. Now, suppose inductively that $(a + b)^2 = a^2 + 2ab + b^2$. Observe:

$$\begin{aligned} \big(S(a) + b\big)^2 &= \big(S(a) + b\big)\big(S(a) + b\big) \\ &= S(a) \times \big(S(a) + b\big) + b \times \big(S(a) + b\big) \\ &= \big(S(a)\big)^2 + S(a) \times b + b \times S(a) + b^2 \\ &= \big(S(a)\big)^2 + S(a) \times b + S(a) \times b + b^2 \\ &= \big(S(a)\big)^2 + 2S(a) \times b + b^2 \end{aligned}$$

□

Problem 5 - Euclidean Algorithm

Let n be a natural number and q be a positive natural number. Then, there exist natural numbers m and r such that $0 \leq r < q$ and $n = mq + r$.

Proof. We induct on n while keeping q fixed. For the base case of $n = 0$, notice that $0 = 0q + 0 = 0 + 0 = 0$. Now, suppose inductively that there exist natural numbers m and r such that $0 \leq r < q$ and $n = mq + r$. We will now show that for $S(n)$, there exist natural numbers m' and r' such that $0 \leq r' < q$ and $n = m'q + r'$. By Problem 2 of Section A, there exists $p \in \mathbb{N}$ such that $S(p) = q$. We now consider two cases of: $r = p$ and $r \neq p$.

1. Say $r = p$. Notice $S(n) = S(mq + r) = S(mq + p) = mq + S(p) = mq + q = (m + 1)q + 0$. Overall, we have that $S(n) = (m + 1)q + 0$.

2. Say instead that $r \neq p$. Notice $S(n) = S(mq + r) = mq + S(r)$. Since $r \neq p$, we have that $r < p$. Otherwise, $r \geq p$. In this case, $r = p$ would cause a contradiction to our original assumption that $r \neq p$. If $r > p$, this would imply that $r \geq S(p) = q$, which break our assumption that $r < q$. As such, $S(r) \leq p < q$. Hence, $S(r) < q$.

In either case, there exist natural numbers m', r' such that $0 \leq r' < q$ and $S(n) = m'q + r'$. □